

ZHONGNING, China

WMO: 537050

Lat: 37.4843N

Lon: 105.6837E

Elev: 1193

StdP: 87.79

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.2	(c) -13.2	(d) -25.1	(e) 0.4	(f) -9.1	(g) -23.2	(h) 0.5	(i) -7.5	(j) 9.0	(k) -0.7	(l) 8.1	(m) -2.8	(n) 2.0	(o) 220	(p) 0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 33.8	(d) 18.8	(e) 32.5	(f) 18.4	(g) 31.1	(h) 18.0	(i) 21.8	(j) 29.5	(k) 21.0	(l) 28.4	(m) 20.2	(n) 27.4	(o) 2.8	(p) 40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	16.6	25.3	18.7	15.6	24.8	17.8	14.7	24.2	70.4	29.6	66.9	28.4	63.9	27.3	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 7.6	(c) 6.5	(e) -18.0	(f) 36.4	(g) 2.4	(h) 1.2	(i) -19.7	(j) 37.3	(k) -21.1	(l) 38.0	(m) -22.5	(n) 38.7	(o) -24.3	(p) 39.6		
(a) 9.0	(b) 7.6	(c) 6.5	(e) -18.7	(f) 23.1	(g) 2.3	(h) 1.1	(i) -20.3	(j) 23.9	(k) -21.7	(l) 24.6	(m) -22.9	(n) 25.2	(o) -24.6	(p) 26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-5.1	-0.7	6.6	13.6	18.6	22.9	24.9	23.2	18.1	11.4	3.8	-3.1		
	DBStd	10.94	3.73	4.56	4.78	4.46	3.69	2.98	3.03	3.15	3.39	3.83	3.85	3.50		
	HDD10.0	1535	468	300	126	16	1	0	0	0	1	29	188	407		
	HDD18.3	3187	726	533	364	151	43	3	1	3	44	217	437	665		
	CDD10.0	1983	0	1	19	125	266	387	461	408	243	72	1	0		
	CDD18.3	593	0	0	0	10	50	140	202	153	37	1	0	0		
	CDH23.3	5759	0	0	8	126	542	1378	2033	1357	295	19	0	0		
	CDH26.7	2159	0	0	1	30	150	523	881	508	66	1	0	0		
(14) Wind	WSAvg	2.8	2.8	2.9	3.3	3.3	3.1	2.8	2.7	2.5	2.5	2.4	2.5	2.7		
(15) Precipitation	PrecAvg	216	1	2	4	12	22	22	41	59	31	16	5	0		
	PrecMax	390	8	10	17	45	68	64	116	160	87	42	23	3		
	PrecMin	83	0	0	0	0	0	2	6	13	2	0	0	0		
	PrecStd	80	2	2	5	12	18	16	29	43	21	11	6	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	15.9	24.1	30.2	31.8	34.4	36.2	34.8	31.4	26.1	18.2	9.9		
		MCWB	1.1	5.3	9.6	13.2	14.6	16.8	19.9	20.2	17.5	13.7	8.3	2.1		
	2%	DB	6.0	11.8	19.9	26.7	30.1	32.9	34.2	32.8	28.5	23.2	15.5	7.2		
		MCWB	-0.8	3.0	7.6	11.7	14.3	16.9	19.6	19.6	16.3	12.4	6.8	0.5		
	5%	DB	3.6	9.4	17.3	24.3	28.4	31.4	32.8	31.2	26.7	20.9	13.2	5.0		
		MCWB	-2.1	1.9	6.6	10.6	13.7	16.7	19.4	19.1	15.8	11.5	5.7	-0.6		
	10%	DB	1.5	7.2	14.9	22.0	26.5	29.8	31.3	29.6	24.8	18.8	10.8	3.1		
		MCWB	-3.5	0.5	5.3	9.7	13.1	16.4	19.2	18.7	15.4	10.3	4.5	-1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	5.6	10.2	14.8	17.5	20.6	23.6	22.9	20.2	14.9	9.1	2.7		
		MCDB	8.6	14.3	21.6	25.2	26.3	28.9	31.4	30.7	25.7	22.9	16.1	8.4		
	2%	WB	-0.6	3.6	8.4	13.1	16.1	19.5	22.1	21.7	18.5	13.6	7.5	0.7		
		MCDB	5.8	10.8	18.2	23.1	25.3	28.0	30.3	29.1	24.7	20.9	13.9	6.3		
	5%	WB	-2.0	2.2	7.1	11.8	15.1	18.7	21.2	20.9	17.4	12.6	6.2	-0.5		
		MCDB	3.4	8.8	16.0	21.8	25.2	27.0	29.1	28.0	23.3	19.0	11.9	4.6		
	10%	WB	-3.4	0.8	5.8	10.6	14.2	17.9	20.4	20.1	16.5	11.4	4.8	-1.7		
		MCDB	1.1	6.9	14.3	19.9	23.8	26.2	28.0	26.9	22.4	17.3	10.3	2.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	12.5	12.5	12.8	12.5	11.8	10.7	10.5	10.8	12.1	11.1	10.6		
		MCDBR	15.0	15.9	16.1	15.9	15.6	14.3	12.8	12.8	13.9	14.7	14.9	13.3		
	5% WB	MCWBR	9.8	9.3	7.8	6.2	5.6	4.5	3.8	4.1	4.7	6.0	7.9	8.4		
		MCWBR	14.8	15.3	14.8	13.1	12.6	11.2	10.3	10.4	10.3	12.9	13.8	12.8		
(40) Clear-Sky Solar Irradiance	taub		0.369	0.430	0.490	0.531	0.534	0.527	0.507	0.522	0.498	0.459	0.395	0.357		
	taud		2.005	1.870	1.768	1.714	1.772	1.850	1.977	1.914	1.912	1.924	1.998	2.060		
	Ebn at Noon		773	769	763	760	767	771	784	760	749	730	736	758		
	Edh at Noon		129	169	207	231	221	205	179	187	177	158	129	115		
(44) All-Sky Solar Radiation	RadAvg		2.79	3.70	4.92	5.87	6.48	6.52	6.17	5.60	4.50	3.97	3.12	2.66		
	RadStd		0.29	0.27	0.26	0.32	0.32	0.39	0.34	0.35	0.34	0.31	0.24	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.54	N/A	+1.10	+0.68	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.42	N/A	N/A	+0.45	N/A	-0.69	N/A	N/A	+74	+40

Nomenclature: See separate page